gdTAI-kimi: an honest-benchmark experiment for the gdTAI program

Date: 2026-08-07 **Status:** experimental candidate, **not registered, not promotable**. gdTAI-kimi is a separately named experiment run outside the frozen GDTAI V4 precommitted protocol (`docs/GDTAI_V4_PRECOMMITTED_PLAN.md`), applying that protocol's core principles end-to-end. Its purpose is twofold: (1) test whether the V4 protocol is executable as written, and (2) quantify how gdTAI-class models behave when every split, feature decision, calibration step, and threshold is grouped, fold-local, and free of adaptive iteration.

One-paragraph answer. Under fully honest nested leave-one-dataset-out evaluation, a from-scratch two-stage gdT classifier (gdTAI-kimi, HistGBM variant) reaches dataset-macro F1 **0.811** (donor-cluster bootstrap 95% CI [0.761, 0.845]) — far above the fairly retrained simple baselines (0.57–0.60) but below the frozen V2/V3 profiles' descriptive scores (~0.91, which are optimistically biased in this frame because those models were developed with access to HRA005041 and GSE144469 cells). **No kimi variant met the precommitted balanced-mode guardrails in any outer fold**, so under the V4 promotion rule gdTAI-kimi would be published as an experimental negative result. The binding constraints were recall at the 0.2% paired-abT FPR ceiling (folds 1 and 3) and the strict-NK FPR ceiling (fold 2). NK leakage — not αβ leakage — is the hardest unsolved problem in this model class.

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1. Design

Element	gdTAI-kimi choice
Evaluation	Nested leave-one-dataset-out over HRA005041, GSE144469, BALF_BLOOD_COPD; 3-fold StratifiedGroupKFold inner tuning, group = donor (fallback sample/library)
Labels	Expression-independent, computed from raw CDR3 fields: `gdT_primary` = paired TRG/TRD, no ab evidence; `abT_primary` = paired TRA/TRB, no gd evidence, **restricted to gdTCR-assayed sublibraries**; censored paired-abT = weight-0.25 training-only weak negatives; sorted-gdT cells without paired TCR = weight-0.25 training-only weak positives; dual/ambiguous excluded
Features	Frozen 197-gene universe from the V4 plan + fold-local derived features (family detection counts, panel means, Stage-1 OOF probability); no Phase-4 scores, no metadata, no annotations
Architecture	Stage 1 soft T-lineage gate (elastic-net; OOF probability is a Stage-2 feature; no cell hard-removed). Stage 2: elastic-net logistic (16-config grid) vs HistGBM (16-config grid), identical folds and weights

Element	gdTAI-kimi choice
Calibration/thresholds	Sigmoid (Platt) on inner-OOF scores; thresholds selected on inner-OOF under the V4 plan's guardrails (balanced: macro recall ≥0.80, paired-abT FPR ≤0.2%, strict-NK FPR ≤1%; high-purity: ≥0.70 / ≤0.1% / ≤0.5%)
Comparators (same folds)	C1 Phase-4 TRD−TRAB score; C2 compact 7-gene logistic; C3 TCR-gene-only elastic-net
Frozen references	V2 high-F1, V2 high-purity, V3 Round 12, V3 Round 14 scored on identical held-out cells — **descriptive only** (see §4.2)
Statistics	Per-dataset metrics primary; 1,000-rep donor-cluster bootstrap CIs on dataset-macro F1

Deviations from the frozen V4 protocol (documented in the script docstring): GSE144469 expression from the integrated-plus6 atlas X; SGD optimizer for elastic-net; fold-local ≤100k tuning subsamples; no extension-cohort guardrails (sealed cohorts stay sealed); no full-atlas inference.

2. Label inventory (all expression-independent)

source	gdT_primary	abT_primary	abT_censored_weak	strict_NK	sorted weak	dual excluded
HRA005041	4,894	22,211	520,817	757	—	5,917
GSE144469	4,003	61,729	2,989	307	—	1,103
BALF_BLOOD_COPD	1,046	33,117	—	254	—	3,116
GDT_2020AUG_woCOV	15,066	—	—	—	10,838	—
MalteGDT	5,650	—	—	—	2,150	—
GSE254249 (stress only)	—	—	174,010	8,198	—	—

Notes: HRA005041 contributes only 22,211 sublibrary-clean abT negatives (vs 520,817 censored) — exactly the M1 trade-off, and it is workable. BALF contributes only 254 strict-NK cells, making per-dataset NK FPR estimates on BALF extremely coarse.

3. Headline results

3.1 Dataset-macro (mean over the three held-out datasets)

candidate	macro F1	bootstrap 95% CI	macro recall	macro precision	macro abT FPR	macro NK FPR	min per-dataset recall
kimi HistGBM balanced	**0.811**	**[0.761, 0.845]**	0.752	0.915	0.63%	5.3%	0.676

candidate	macro F1	bootstrap 95% CI	macro recall	macro precision	macro abT FPR	macro NK FPR	min per-dataset recall
kimi elastic-net balanced	0.634	[0.605, 0.655]	0.591	0.937	0.47%	6.9%	**0.083**
kimi HistGBM high-purity	0.727	—	0.630	0.939	0.43%	4.1%	0.469
C1 Phase-4 TRD−TRAB	0.573	[0.512, 0.607]	0.477	0.929	0.44%	1.6%	0.211
C2 compact 7-gene	0.595	[0.497, 0.639]	0.515	0.923	0.57%	2.8%	0.208
C3 TCR-gene elastic-net	0.582	[0.471, 0.628]	0.495	0.936	0.45%	2.0%	0.209
frozen V2 high-purity*	0.906	[0.891, 0.916]	0.885	0.930	0.40%	4.6%	0.829
frozen V3 Round 12*	0.913	[0.888, 0.922]	0.888	0.944	0.36%	8.1%	0.821
frozen V3 Round 14*	0.910	[0.897, 0.917]	0.886	0.937	0.38%	7.8%	0.831

*Frozen profiles are descriptive references. They were trained/selected with access to HRA005041 and GSE144469 cells, so their numbers on those datasets are train-adjacent and optimistically biased; see §4.2.

3.2 Per held-out dataset (balanced mode)

held-out	candidate	precision	recall	F1	abT FPR	NK FPR	ROC-AUC
HRA005041	kimi HistGBM	0.999	0.677	0.807	0.02%	14.9%	0.998
HRA005041	kimi elastic-net	0.980	0.811	0.887	0.36%	20.0%	0.991
HRA005041	frozen V3 R14*	0.998	0.962	0.979	0.05%	22.1%	0.999
GSE144469	kimi HistGBM	0.760	**0.902**	0.825	1.85%	1.0%	0.970
GSE144469	kimi elastic-net	0.843	0.879	0.861	1.06%	0.7%	0.948
GSE144469	frozen V3 R14*	0.850	0.866	0.858	0.99%	1.0%	0.962
BALF_BLOOD_COPD	kimi HistGBM	0.986	0.676	0.802	0.03%	0.0%	0.980
BALF_BLOOD_COPD	kimi elastic-net	0.989	**0.083**	0.153	0.00%	0.0%	0.980

held-out	candidate	precision	recall	F1	abT FPR	NK FPR	ROC-AUC
BALF_BLOOD_COPD	frozen V3 R14*	0.963	0.831	0.892	0.10%	0.4%	0.969

4. Findings

4.1 Kimi fails the precommitted guardrails in every fold — an honest negative result

No kimi variant satisfied the balanced-mode guardrails during inner tuning. The binding constraint differed by fold: training on GSE144469+BALF, the best achievable macro recall at the 0.2% abT-FPR ceiling was 0.66 (HistGBM); training on HRA005041+GSE144469, elastic-net reached only 0.40; training on HRA005041+BALF, recall was fine (0.95) but the inner strict-NK FPR was 14–20%, 14–20× over the 1% ceiling. Under the V4 promotion rule this experiment ends here: V3 Round 14 stays the default and kimi is published as a negative result. This is the protocol working as intended.

4.2 The frozen models' advantage is real but smaller than their headline numbers imply

The only like-for-like frame is BALF_BLOOD_COPD (never used to fit any model; used in frozen model *selection*, so still mildly favorable to them). There, frozen V3 R14 beats kimi HistGBM on recall (0.831 vs 0.676) at comparable precision. On HRA005041 and GSE144469 the frozen models' numbers are train-adjacent: V3 trained on both datasets' cells, so its 0.96-recall on "held-out" HRA005041 is not a generalization measurement. Notably, on GSE144469 — where frozen V3 had every advantage — kimi HistGBM still achieved *higher* recall (0.902 vs 0.866), at the cost of a higher abT FPR (1.85% vs 0.99%).

4.3 NK leakage is the binding constraint, not αβ leakage

Across folds and model families, the strict-NK FPR is 5–50× harder to control than the paired-abT FPR (e.g., kimi HistGBM on held-out HRA005041: abT FPR 0.02% vs NK FPR 14.9%). This is not unique to kimi: frozen V3 R14's NK FPR on the same cells is 22.1% — *worse* than kimi HistGBM. The historical NK hard-negative strategy (v3's TRDC+TRDV− guard) addressed a visible symptom; the grouped, never-fitted NK evaluation shows the problem is deeper. The V4 protocol's soft-gate-only NK defense is likely insufficient as written; see §5.

4.4 gdTAI-class models add large, real value over simple baselines

The fairly retrained comparators collapse under honest evaluation: macro F1 0.57–0.60, with per-dataset recall as low as 0.21 on BALF. The full two-stage model roughly doubles comparator recall at similar precision. The answer to "is a TRD−TRAB score good enough" is clearly no — but the answer to "do we need more than ~200 genes and a small gradient-boosted tree" is also no.

4.5 Elastic-net is unstable across datasets; HistGBM is robust

The elastic-net variant collapsed on held-out BALF (recall 0.083 balanced, 0.027 high-purity) while performing acceptably on the other two folds — its learned linear weights do not transfer across chemistries/tissues. HistGBM's worst-dataset recall was 0.676. Family choice matters less than this stability gap; any future protocol should score worst-dataset recall, not just macro.

4.6 Fold-to-fold model instability is itself a finding

Different folds selected different configs (en_a0.001_l0.75 / en_a0.0003_l0.25 / en_a0.001_l0.5; hgb_14 / hgb_12 / …) and the GSE254249 stress FPR varied >10× across fold models (0.01%–1.3% for HistGBM balanced). A single "final fit on all data" would hide this variance; the V4 protocol's feature-stability promotion condition (§6.3 item 5) is well-motivated and should be extended to threshold stability.

4.7 CD4/Treg post-exclusion is nearly free

The frozen CD4-helper/Treg exclusion rules cost only 1–20 gold-positive cells per dataset (≤0.5% recall) while providing a deterministic safety rail. Keep them.

5. Implications for the frozen V4 protocol

1. **Guardrail feasibility risk is real.** The 0.80 macro-recall at ≤0.2% abT-FPR balanced guardrail was unreachable in two of three folds for a fresh model family. Before Step 2, the preflight should verify the guardrails are jointly satisfiable on inner-OOF predictions, or the V4 run will end in a guaranteed guardrail failure.
2. **Add an NK defense to Stage 2, or accept NK FPR as the limiting metric.** Options: include never-evaluated NK hard negatives in Stage-2 training (v3's approach, but with grouped folds and representative NK sampling rather than TRDC+TRDV−-only), or make the Stage-1 gate firmer for NK-like cells (a soft gate lets NK leakage through by design).
3. **Report guardrail-failed candidates anyway.** Kimi's per-dataset numbers remain informative precisely because the failure is flagged, not hidden. The V4 report template should include best-effort rows with `guardrails_met=false`.
4. **Adopt worst-dataset recall as a primary metric** (§4.5) and threshold-stability reporting (§4.6).
5. **BALF's 254 strict-NK cells cannot support a 1% NK-FPR guardrail at dataset level** (1% = 2.5 cells). NK guardrails should be pooled across datasets or BALF's NK stratum acknowledged as uninformative.

6. Caveats

- gdTAI-kimi used the atlas X matrix for GSE144469 rather than the raw SRR-joined object the V4 plan mandates; results may differ slightly under the plan's data contract.
- Labels remain TCR-pipeline-derived (review finding F2): kimi measures TCR-call reproduction from RNA, same as its predecessors. The two-tier negative scheme controls *assay censoring* bias, not label circularity.
- Frozen-profile numbers are descriptive and biased in their favor on atlas datasets; the honest kimi-vs-frozen comparison is BALF only.
- Extension-cohort guardrails and full-atlas inference were intentionally not run (sealed cohorts remain sealed; whole-atlas inference requires separate approval).

8. Round 2: diagnosis-driven iteration (2026-08-07, same day)

Round 1 showed three failure drivers: NK leakage as the binding constraint, BALF positives clustering just below the transferred threshold, and HGB grid capacity limits. Round 2 made exactly four pre-registered changes: **R2-1** grouped NK hard negatives (weight 0.5) added to Stage-2 training; **R2-2** TRD-only silver-like cells added as weight-0.25 weak positives; **R2-3** HistGBM capacity raised (leaves 15–31, max_iter 400); **R2-4** recall-at-fixed-FPR operating points reported for every candidate.

8.1 Round 1 vs round 2 (dataset-macro, balanced mode)

candidate	macro F1 r1 → r2	macro recall r1 → r2	macro abT FPR r1 → r2	macro NK FPR r1 → r2
kimi HistGBM balanced	0.811 → **0.838**	0.752 → **0.828**	0.63% → 1.05%	5.3% → 17.1%
kimi HistGBM high-purity	0.727 → 0.789	0.630 → 0.720	0.43% → 0.61%	4.1% → 6.4%
kimi elastic-net balanced	0.634 → 0.464	0.591 → 0.419	0.47% → 0.79%	6.9% → 1.2%
C2 compact 7-gene	0.595 → 0.589	—	—	—

Round 2 bought recall but paid for it in false positives — it moved along the recall/FPR frontier rather than dominating round 1. The silver weak positives (R2-2) broadened the positive class but pulled TRD-positive NK-like cells with them (GSE144469 held-out NK FPR rose 1.0% → 32.9%): **TRD-only evidence cannot distinguish silver gdT from NK contamination, and adding silver to fitting was counterproductive for NK control.** This independently confirms protocol v1.2's decision to remove silver and sorted cells from fitting. The NK training negatives (R2-1) did not reduce NK FPR either (fold-2 inner NK FPR ~20%, unchanged) — NK leakage in this feature space is not a sample-size problem.

8.2 The useful output: the honest recall@FPR frontier (kimi HistGBM, round 2)

Thresholds set on inner-OOF to fixed abT-FPR targets, applied to the held-out dataset:

held-out	recall @ ≤0.1% target	recall @ ≤0.2% target	recall @ ≤0.5% target	recall @ ≤1% target
HRA005041	0.627 (meas. 0.01%)	0.793 (0.02%)	0.952 (0.20%)	0.983 (1.24%)
GSE144469	0.910 (meas. 1.87%)	0.914 (3.06%)	0.921 (7.28%)	0.951 (14.95%)
BALF_BLOOD_COPD	0.487 (meas. 0.01%)	0.624 (0.02%)	0.841 (0.13%)	0.926 (1.46%)

Two transferable lessons: (a) thresholds calibrated on atlas sources transfer well to HRA005041 and BALF (measured FPR tracks the target) but **fail badly toward GSE144469** (measured FPR up to 19× the target) — a single global threshold is not robust across that dataset's direction; (b) the honest achievable operating point is roughly **macro recall ≈ 0.90 at 0.5% abT FPR** (0.95 / 0.92 / 0.84 per dataset), not the 0.80 recall at 0.2% FPR the balanced guardrail demands — the precommitted guardrails sit slightly off the achievable frontier for any model in this family, which is exactly what the parallel V4 v1.2 CPU run found independently.

8.3 Round-2 conclusion

No round-2 variant met the guardrails; kimi remains an experimental negative result, now with a mapped frontier. The evidence-based guidance for the V4.1-GPU plan: keep silver and sorted cells out of fitting; treat NK FPR as the primary limiting metric and pool NK guardrails across datasets (BALF has 254 strict-NK cells); report recall@fixed-FPR curves instead of relying on one transferred threshold; and expect the balanced guardrail (recall ≥0.80 at ≤0.2% FPR) to be reachable only marginally, if at all, without a structurally different NK defense.

9. Round 3: NK label repair (2026-08-08)

Round 3 tested the NK-label-contamination hypothesis directly, with two pre-registered changes: **R3-1** strict-NK negatives restricted to gdTCR-assayed sublibraries (cells whose "no gdTCR" is censoring rather than evidence are excluded from fitting and from FPR denominators); **R3-2** NK-annotated cells with TRDC expression split out as an abstain-zone diagnostic. Silver weak positives were reverted (R2-2 was counterproductive); NK hard negatives in Stage-2 training and the larger HGB grid were retained.

The repair was drastic, as predicted from the assay audit: HRA005041 strict NK dropped 757 → **15** (98% of its NK "negatives" were in sublibraries that could not see γδ chains); GSE254249's entire 8,198-cell NK stratum is censored and unusable; BALF dropped 254 → 214 plus 40 TRDC-ambiguous. Total clean NK across the three LODO sources: **533 cells**.

9.1 The headline finding: the "NK leakage" was mostly the measuring instrument

With clean NK denominators, **strict-NK FPR is 0.000 for every model on every dataset** — kimi elastic-net, kimi HistGBM, and the frozen V2/V3 profiles alike (single exception: frozen V3 R14 calls 1 of 214 clean BALF NK cells). Round 1's alarming 15–22% NK FPR on HRA005041 was measured almost entirely against the 741 censored cells; on the 15 cells where NK status is actually observable, no model calls any of them positive. The censored-NK call rate remains 18–23% for all models — a number that cannot be interpreted as model error without gdTCR assays for those cells, and which bounds the true NK FPR from above.

Two consequences:

1. **gdTAI v3's NK-guard architecture was solving a label problem with model capacity.** The frozen models' NK FPR also collapses to ~0 on clean denominators — they were never as NK-leaky as the contaminated metric claimed.
2. **The NK guardrail as currently defined is unmeasurable, not failed.** 533 clean NK cells (15/304/214 per dataset) cannot support a 1% per-dataset FPR guardrail. NK evaluation must be pooled across datasets and reported with Wilson intervals, and the censored-NK call rate must be reported separately as an upper bound.

9.2 Round-3 metrics (balanced mode)

candidate	macro F1 (95% CI)	macro recall	macro precision	macro abT FPR	clean NK FPR	inner guardrails
kimi HistGBM	0.831 [0.794, 0.857]	0.783	0.911	0.65%	0.000	met in fold 2
kimi elastic-net	0.551 [0.439, 0.600]	0.465	0.952	0.28%	0.000	met in fold 2
frozen V3 R14 (descriptive)	~0.91	0.886	0.937	0.38%	0.000–0.5%	n/a

Per held-out dataset (kimi HistGBM): HRA005041 recall 0.785 / abT FPR 0.03%; GSE144469 0.904 / 1.88%; BALF 0.662 / 0.05%. Both families passed the full balanced guardrail set in inner tuning for fold 2 (recall 0.85/0.95, abT FPR ≤0.2%, NK FPR 0) — the first guardrail pass of the kimi experiment, enabled by honest NK labels. Fold 1 and fold 3 still fail on recall at the 0.2% FPR ceiling (inner recall 0.73 and 0.71 for HistGBM), and the BALF recall gap (~0.66 vs frozen 0.83) is unaffected by the NK repair — that gap is real model difference on TCR-defined positives, not label noise.

9.3 What round 3 settles

- The NK problem was **two problems**: a contaminated measuring instrument (now fixed — and the fix must be ported to the V4 protocol's NK definition and the extension-screen NK strata) and a smaller genuine model-error component bounded by the censored-NK call rate (18–23% on unverifiable cells; unresolvable without gdTCR assays — an abstain zone is the honest answer).
- The remaining performance gap is concentrated in **recall at strict FPR** (folds 1/3) and **BALF cross-study recall** — neither is NK-related. BALF positives remain biologically hard (tissue Vδ1-like cells under different chemistry); without new gdTCR data this is close to the irreducible floor for a 197-gene RNA-only model.
- The elastic-net family's instability (BALF recall 0.09) persists across all three rounds; HistGBM is the only viable Stage-2 family in this program.

7. Artifacts

- Script: `workflows/gdtai/run_gdtai_kimi_nested.py` (seed 20260807)
- Tables: `Integrated_dataset/tables/gdT_prediction/gdtai_kimi/` (per-dataset and macro metrics, per-candidate inner diagnostics, bootstrap distributions, label inventory, per-cell held-out scores)
- Figures: `Integrated_dataset/figures/gdT_prediction/gdtai_kimi/`
- Experimental fold models: `Integrated_dataset/models/gdT_prediction_classifier/gdtai_kimi/` (explicitly unregistered, not promotable)
- Run log: `Integrated_dataset/logs/gdT_prediction/gdtai_kimi/run.log`